ApexSphere Q1 Data Center Operations Report

Prepared by Site Operations · Reporting period: January–March · Classification: Internal (fictional)

1. Executive summary

The Atlanta campus (ATL-1) ran at an average utilization of 87% of critical IT capacity during Q1, well above the 75% operating target set by the Infrastructure Committee. Sustained operation above target reduces the headroom available for maintenance windows and failover.

Availability remained strong at 99.982% across all customer-facing halls. Power usage effectiveness (PUE) averaged 1.48 against a target of 1.35, driven mainly by the age of the ATL-1 cooling plant.

Site Operations recommends that leadership treat ATL-1 capacity and resilience as a single decision: new demand cannot be absorbed safely until headroom is restored.

Metric	Q1 actual	Target
ATL-1 utilization	87%	75%
Availability	99.982%	99.98%
PUE	1.48	1.35

2. Capacity and demand

ATL-1 has 18 MW of critical IT capacity, of which 15.7 MW was in use at quarter end. The Dallas campus (DAL-2) ran at 61% utilization and has 4.1 MW of unsold capacity, but most Atlanta customers require low-latency connectivity to the Southeast metro and cannot be relocated to Dallas.

The sales team holds a customer waitlist of 2.4 MW for ATL-1, including two hyperscale tenants. Without additional capacity, roughly 1.1 MW of that demand is expected to go to competitors within two quarters.

3. Incidents and staffing

There were three Severity-2 incidents in Q1 and no Severity-1 incidents. The most significant occurred on February 11, when UPS Bank B at ATL-1 failed over unexpectedly and Hall 3 experienced 14 minutes of degraded cooling. No customer workloads went offline, but two tenants filed formal inquiries.

Mean time to restore (MTTR) across all incidents was 3.2 hours. ATL-1 currently has 6 open critical facilities technician positions, and overtime for the facilities team rose 22% compared with Q4.

4. Cooling and sustainability

Chiller Plant 2 at ATL-1 was installed in 2009 and has reached the end of its rated service life. Its failure would remove about 30% of the campus's cooling capacity during summer peaks.

Water usage effectiveness (WUE) averaged 1.9 liters per kWh. The sustainability team expects that replacing Chiller Plant 2 would lower ATL-1 PUE to about 1.38.